Inductive proof of Nicomachus's Theorem

Heejun Yoon

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Abstract

This paper shows proof by induction of Nicomachus's Theorem in high school level mathematics.

Contents

1 Introduction

While Nicomachus's theorem is a classical result in number theory, it may be applied for efficient calculation for computers.

2 Nicomachus's Theorem

Nicomachus's Theorem shows that the addition of $1^3, 2^3 \dots n^3$ is identical to the addition of 1 to n squared. This is written as:

$$\sum_{i=1}^{n} i^3 = \left(\sum_{i=1}^{n} i\right)^2 \tag{1}$$

3 Induction

The statement $P(n)$ is defined as for $n \in \mathbb{Z}^+$,

$$\sum_{i=1}^{n} i^3 = \left(\sum_{i=1}^{n} i\right)^2 \tag{2}$$

3.1 Base case

when $n = 1$,

$$\begin{aligned} \text{LHS} &= \sum_{i=1}^{1} i^3 \\ &= 1^3 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= \left(\sum_{i=1}^{1} i\right)^2 \\ &= (1)^2 \\ &= 1 \end{aligned}$$

Therefore, LHS = RHS, base case $n = 1$ holds.

3.2 Inductive Hypothesis

Assume the statement $P(k)$, for $k \in \mathbb{Z}^+$,

$$\sum_{i=1}^{k} i^3 = \left(\sum_{i=1}^{k} i \right)^2 \tag{3}$$

holds.

3.3 Proof by induction

Prove that the statement $P(k+1)$

$$\sum_{i=1}^{k+1} i^3 = \left(\sum_{i=1}^{k+1} i \right)^2 \tag{4}$$

holds.

$$
\begin{aligned}
\text{RHS} &= \left(\sum_{i=1}^{k+1} i\right)^2 \\
&= \left\{\sum_{i=1}^{k} i + (k+1)\right\}^2 \\
&= \left(\sum_{i=1}^{k} i\right)^2 + 2(k+1) \times \sum_{i=1}^{k} i + (k+1)^2 \\
&= \sum_{i=1}^{k} i^3 + (k+1)\left(k+1+2\times\sum_{i=1}^{k} i\right) && \text{by 3.2} \\
&= \sum_{i=1}^{k} i^3 + (k+1)\left\{k+1+2\times\left(\frac{k(k+1)}{2}\right)\right\} && \text{by sum of arithmetic sequence} \\
&= \sum_{i=1}^{k} i^3 + (k+1)(k+1+k^2+k) \\
&= \sum_{i=1}^{k} i^3 + (k+1)(k^2+2k+1) \\
&= \sum_{i=1}^{k} i^3 + (k+1)(k+1)^2 \\
&= \sum_{i=1}^{k} i^3 + (k+1)^3 \\
&= \sum_{i=1}^{k+1} i^3
\end{aligned}
$$

$\therefore \text{LHS} = \text{RHS}$ $\square$

4 Conclusion

Hence, according to the Principle of Mathematical Induction, since the base case and the inductive step are both proven, the statement holds for all $n \in \mathbb{Z}^+$.